

Computational Biomathematics: Pattern Formation & Reaction-Diffusion Systems

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Pattern formation shows that simple local interactions can generate complex global structures. Understanding these mechanisms is central to mathematical biology.

In this lab, we study how spatial patterns such as spots, stripes, and waves can emerge from simple local interactions. This phenomenon is known as pattern formation and is closely related to the work of Alan Turing.

The key idea is that two interacting substances (often called morphogens) can form patterns when they react with each other and diffuse through space at different rates.

Complex biological patterns can arise from simple local rules involving reaction and diffusion.

Examples include:

- animal skin patterns (zebra stripes, leopard spots),
- chemical reactions,
- developmental processes in embryos.

General Model Idea

We consider two variables:

- $u(x, y, t)$: activator
- $v(x, y, t)$: inhibitor

Their dynamics follow the general structure:

<pre>u_next = u + (Du * laplacian(u) + f(u,v)) * dt v_next = v + (Dv * laplacian(v) + g(u,v)) * dt</pre>
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You do not need to derive these equations. Your task is to **experiment with them and interpret the results**.

A Concrete Reaction–Diffusion Model

In this lab, we use a specific reaction–diffusion system known as the Gray–Scott model. This model is widely used to study pattern formation.

The equations are given by

$$\begin{aligned}\frac{\partial u}{\partial t} &= D_u \Delta u - uv^2 + F(1 - u), \\ \frac{\partial v}{\partial t} &= D_v \Delta v + uv^2 - (F + k)v.\end{aligned}$$

Here:

- $u(x, y, t)$ represents a background chemical (nutrient),
- $v(x, y, t)$ represents an activator,
- D_u, D_v are diffusion rates,
- F is a feed rate,
- k is a decay rate.

In discrete form, the update rule becomes:

```
u = u + (Du * laplacian(u) - u * v**2 + F * (1 - u)) * dt
v = v + (Dv * laplacian(v) + u * v**2 - (F + k) * v) * dt
```

A suitable initial condition is to start with $u = 1$ and $v = 0$ everywhere, and introduce a small disturbance in the center of the domain.

Hint: Different choices of the parameters F and k lead to very different patterns such as spots, stripes, or homogeneous states.

Starter Code

You may use the following basic components:

```
import numpy as np
import matplotlib.pyplot as plt

def laplacian(Z):
    return (
        -4*Z
        + np.roll(Z, 1, axis=0)
        + np.roll(Z, -1, axis=0)
        + np.roll(Z, 1, axis=1)
        + np.roll(Z, -1, axis=1)
    )
```

Hint: The Laplacian describes diffusion. You do not need to modify this function.

General Simulation Loop

```
for t in range(T):
    u = u + (Du * laplacian(u) + f(u,v)) * dt
    v = v + (Dv * laplacian(v) + g(u,v)) * dt
```

Working Strategy

First understand your task and what you want to explore. Then formulate a clear AI prompt to generate code. After running the code, interpret the resulting patterns biologically. Each group focuses on a different aspect of pattern formation.

Group A: Diffusion Balance

In this task, you investigate how the relative diffusion rates influence pattern formation. Biologically, one substance spreads quickly (inhibitor), while the other spreads slowly (activator).

Student Task

Run simulations with different values of D_u and D_v . Compare cases where one diffuses much faster than the other and cases where both diffuse at similar rates. Observe how the resulting patterns change.

Hint: Typical pattern formation requires that the inhibitor diffuses faster than the activator.

AI Prompt Design Task

Write a prompt asking an AI system to implement a reaction–diffusion model where diffusion parameters can easily be changed and visualized.

Student Task

Explain why different diffusion rates are essential for pattern formation.

Group B: From Noise to Pattern

In real biological systems, patterns often arise from small random fluctuations rather than pre-existing structures.

Student Task

Experiment with different initial conditions:

- random noise,
- a single localized disturbance,
- structured initial conditions.

Observe how patterns emerge over time.

Hint: Even very small random differences can grow into large-scale structures.

AI Prompt Design Task

Write a prompt asking an AI system to include flexible initialization options and reproducibility (e.g., random seeds).

Student Task

Explain why patterns can emerge from noise.

Group C: Parameter Sensitivity

Biological systems must operate within certain parameter ranges to produce stable patterns.

Student Task

Vary reaction parameters in your model and observe how the system behaves. Try to identify:

- parameter values that produce stable patterns,
- parameter values that destroy patterns.

Hint: Small parameter changes can lead to very different outcomes.

AI Prompt Design Task

Write a prompt asking an AI system to run simulations for multiple parameter sets and visualize the results side by side.

Student Task

Explain why pattern formation only occurs in certain parameter regimes.

Group D: Biological Interpretation

In this task, you connect mathematical patterns to real biological systems.

Student Task

Run the simulation and generate visually clear patterns. Classify the patterns you observe:

- spots,
- stripes,
- waves.

Relate these patterns to biological examples such as animal skin or developmental processes.

Hint: Focus on interpretation rather than technical complexity. The goal is to understand how mathematical patterns relate to biology.

AI Prompt Design Task

Write a prompt asking an AI system to generate clear visualizations and allow parameter tuning to produce different types of patterns.

Student Task

Explain which biological systems could correspond to the patterns you observe.

Reflection

Student Task

Reflect on your work. Did your prompt clearly describe the model? Did the AI-generated code match your expectations? What difficulties did you encounter?

Final Presentations

Each group presents:

- what they varied,
- what patterns emerged,
- their best AI prompt,
- one biological interpretation.